

Marvel's Spider-Man: Miles Morales Procedural Tools for PlayStation 5 Content Authoring

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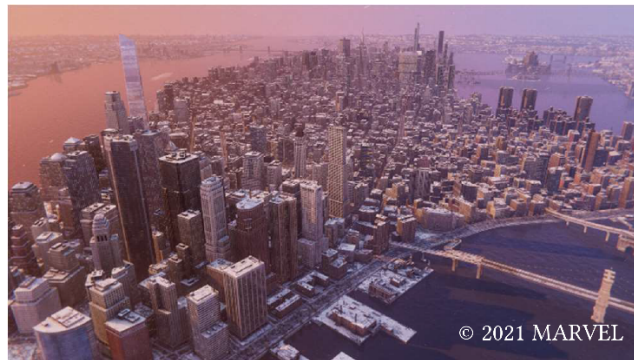


Figure 1: Marvel's Manhattan in winter, captured on PlayStation 5. Various procedural tools combined to help artists generate a city full of content.

ABSTRACT

Artists at Insomniac Games created a wonderful and detailed open world, Marvel's Manhattan in autumn, on the PlayStation 4 console for Marvel's Spider-Man. This technical communication provides an overview of several procedural systems developed or improved upon for the standalone game, Marvel's Spider-Man: Miles Morales, and a snapshot of the procedural processes in use during the game's production on the PlayStation 5 console. Procedural systems allowed artists at Insomniac Games to efficiently update the setting to winter, increase visual fidelity, and propagate mark-up data across a large open world environment. The combination of procedural techniques and bespoke artistry enabled Insomniac Games to create a memorable launch title for the PlayStation 5.

CCS CONCEPTS

• **Software and its engineering** → Software organization and properties; Contextual software domains; Virtual worlds software; Interactive games; • **Computing methodologies** → Computer graphics.

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KEYWORDS

gaming, open world, PlayStation, Spider-Man, Miles Morales, Sony, geometry, procedural, Houdini

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1 INTRODUCTION

The open world environment players explore and traverse in the video game *Marvel's Spider-Man: Miles Morales* required several new features and offline systems to generate content and improve visual fidelity for next generation console hardware when comparing it against the original game, *Marvel's Spider-Man*. Setting *Miles Morales* narrative during wintertime meant adding snow, ice, and holiday decorations across the city [Figure 1]. New enemy types with the ability to super-jump required special mark up for traversal and navigation. The PlayStation 5 supports hardware accelerated ray tracing which necessitated visual fidelity improvements for the IG-Impostor system [Halperin et al., 2018] introduced in *Marvel's Spider-Man* on the PlayStation 4.

Migrating the art and assets that make up Marvel's Manhattan from the PlayStation 4 to the PlayStation 5 required procedural systems to enable a small team of environment artists to manage the content creation pipeline for a large open world environment that measures about six kilometers by nine kilometers and contains thousands of buildings. Introducing dependency graphs into the artist workflow increased iteration speed.

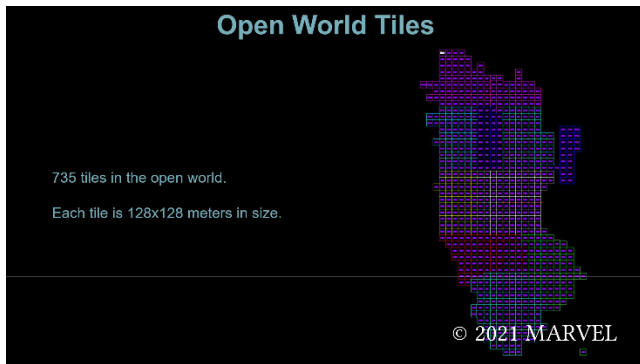


Figure 2: The open world of *Marvel's Spider-Man: Miles Morales* was divided into 735 tile regions, each measuring 128-meters on a side.

2 DEPENDENCY GRAPHS

The use of dependency graphs is new for production at Insomniac Games. Parallel processing using dependency graphs to iterate over game data improved performance of off-line procedural systems by an order of magnitude when comparing the original game to its *Miles Morales*. Existing systems were updated to take advantage of dependency graphs and new systems were authored with dependency graphs in mind.

The data structure for the open world in both *Marvel's Spider-Man* and *Marvel's Spider-Man: Miles Morales* were similar. The open world environment was divided into square “tile regions” measuring 128-meters on a side [Figure 2]. On the first game procedural tools iterated over the data in a serial fashion however during production of *Miles Morales*, all procedural tools utilized dependency graphs to process the tile regions in parallel as work items.

3 WINTERIZING MANHATTAN

Marvel's Spider-Man was set in the Fall, while *Marvel's Spider-Man: Miles Morales* is set during the winter. Several types of snow were authored and added to the open world environment.

3.1 Snow Geometry

Procedural tools iterated over the terrain of the city, measuring the curbs of the city streets. Using this data, the system placed snow mounds on metrics of 2 meters, 4 meters, 8 meters and 16 meters while avoiding driveways and corners by testing against bounding volumes. A similar system placed snow mounds on the lane lines running down the streets, as well as snow skirts around the bases of buildings [Figure 3].

Road, curb, and skirt snow are prefabricated containers made of multiple model instances all built to the same metrics of 2 meters, 4 meters, 8 meters and 16 meters. The procedural snow mound placement system automatically selects variants of prefabs that fit the measured optimized metrics obtained by analyzing the open world terrain while avoiding collisions with curbs, subway stations, bus stops, and other urban features. Instanting the different snow assets as appropriate, matching the metrics of the measured curb, lane line or building skirt target to the centroid of each prefab,



Figure 3: Snow mounds on curbs, lane lines, and building skirts are geometric while vertex colors control snow and ice patterns on sidewalks and asphalt.

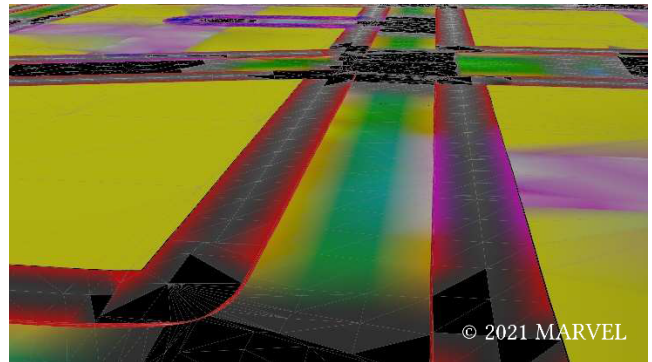


Figure 4: Procedurally generated vertex colors on terrain geometry drive shader components such as ice buildup on sidewalks and directionality of ice streaks on asphalt.

lengthwise along the axis of the measured line segment, assures proper placement of the snow assets.

4 ICE SHADERS AND VERTEX COLORS

Vertex colors procedurally added to the terrain geometry [Figure 4] drove shader components such as linear or omni-directional pattern blending for asphalt ice streaks, and sidewalk icing. The red value on sidewalks controlled where ice builds up around pedestrian paths. Yellow and green vertex colors controlled similar blends for icy asphalt. Alpha values at the intersections of streets controlled the blend of directionality of the ice streaks along roads to a more omni-directional pattern in the middle of intersections.

Procedural systems created vertex colors by analyzing the terrain. Loading in the curves for pedestrian paths, lane lines, and sidewalk edges while super-imposing them against the terrain allowed for attribute transfers from one geometry set to the other. The transferred attributes in turn drove the generation of vertex color attributes on the terrain. The materials assigned to the terrain interpreted the vertex color geometry attributes to further winterize Marvel's Manhattan.

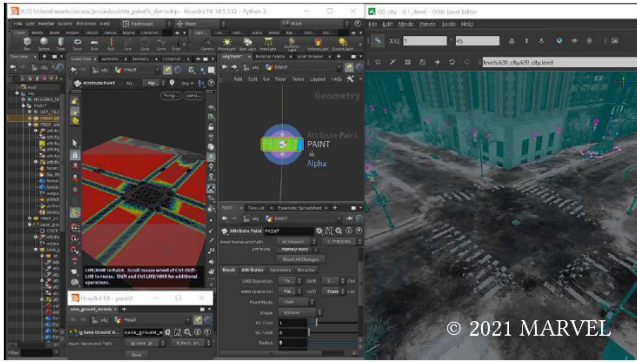


Figure 5: Tight integration between Houdini's vertex paint controls and the Insomniac Level Editor



Figure 6: Procedurally placed holiday decorations.

A tight integration with the vertex paint operations within the Houdini digital content creation (DCC) application and the Insomniac Orbit level editor allowed for near real time integration and visualization. As an artist painted vertex colors in the DCC and saved out the geometry with the adjusted vertex colors, the Insomniac build system detected the updated files and built binary data which was immediately loaded into the Orbit level editor to show the result. [Figure 5]

4.1 Decoration

In *Marvel's Spider-Man: Miles Morales*, winter holiday decorations such as fairy lights on building windows were procedurally populated across Marvel's Manhattan. Variants of window fairy lights were hand modeled and added to a defined asset collection. Matching the fairy light model asset name in the collection with the window model asset name contained in the building assets allowed procedural placement of the fairy light assets onto the windows of each candidate building.

Generating point clouds from the centroids of each model instance from each building to its neighbors provided a base for constructing plausible connections between structures. Power cables, some with holiday decorations [Figure 6], were procedurally placed across various streets throughout the open world of Marvel's Manhattan. Each cable is an asset containing the cable and two

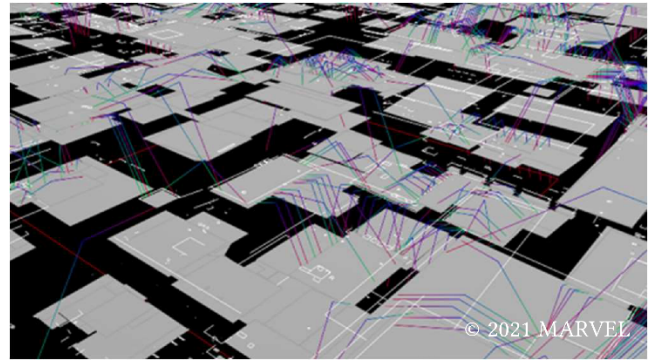


Figure 7: Culled navigation clues, represented as colored lines, propagated to a tile region and its neighbors.

wall mounted attachment points. Attributes from the point clouds of each building and its neighbors provided a base for constructing plausible connections between structures.

Using a procedural system to adjust the attachment point positions for each instance of the cable prefab while adjusting the scale transformation of the geometry of the cable allowed for re-use of a small number of transformed model instances. Analyzing the point clouds allowed the cables to avoid collisions with architectural features such as fire escapes, while material properties of the model geometry referenced by the point cloud provided culling information for implausible connection points such as those upon glass windows.

5 NAVIGATION CLUES

Marvel's Spider-Man: Miles Morales introduced a new enemy, The Underground. These tech-powered foes jump from rooftop to rooftop chasing Spider-Man with 3D pathfinding assisted by procedurally generated navigation clue markup [Figure 7]. The Insomniac Engine uses the markup to inform custom-cost functions for A* pathfinding [Hart et al., 1968]. Each building in the open world environment contains ledge markup indicating the edges of rooftops where Spider-Man may perch. By extracting the roof geometry for each building and combining it with the ledge markup, procedural tools generate a set of possible navigation clues for valid jumps from roof-to-roof and roof-to-ground using baseline heuristics such as vertical and horizontal minimum and maximum distance, and maximum angle using a dot product. After propagating a set of candidate path another set of rules culls the candidate paths using additional heuristics such as:

- Distance
- Ray test for collision along the path
- If a nearby roof is directly above a landing point

There are separate rules for paths that go from roof to roof in the same building vs. those that cross over a street or alley. Additional rules control ground to roof navigation clues.

While iterating over each tile region in the game and taking neighboring tile regions into consideration, the navigation clue system links one tile region to another with navigation clues. Performing analysis of rooftops, ledges, and terrain, the procedural



Figure 8: Ray traced reflections for buildings as close as across the street from the player are IG-Impostors.

system was able to fill Marvel’s Manhattan with an easily updatable and plausible set of navigation clues for The Underground combat scenarios wherever they occur in the open world environment.

6 IG-IMPOSTOR IMPROVEMENTS

The IG-Impostor system developed for *Marvel’s Spider-Man* underwent significant improvement during the production of *Marvel’s Spider-Man: Miles Morales*. IG-Impostors represent the lowest LOD of buildings in the game’s open world. They never unload out of global resident memory. The memory footprint of the geometry assets for Marvel’s Manhattan in *Marvel’s Spider-Man: Miles Morales* was about 100 megabytes. The lowest MIP of IG-impostor textures took up about 400 megabytes.

Insomniac Games leveraged the fact that impostors persist in resident memory and are simple in geometric complexity, as well as rendering cost, to generate the bounding volume hierarchy (BVH) trees for the Insomniac Engine’s extensive use of always-on hardware accelerated ray traced reflections on the PlayStation 5. This new usage case necessitated an across-the-board improvement in the visual quality of IG-Impostors. In *Marvel’s Spider-Man* on the PlayStation 4, the IG-impostors were never seen from closer than about 192 meters.

While most high-resolution environment assets are specifically tagged as invisible to ray traced reflection rays for performance reasons, the IG-Impostors are always visible in the ray-traced reflections while playing *Marvel’s Spider-Man: Miles Morales* and thus appear much closer to the viewer than anticipated for prior games [figure 8].

Individual IG-Impostor texture atlases for the PlayStation 5 were captured at a resolution of 12 to 14 pixels per meter while the PlayStation 4 texture atlases were captured at 8 to 10 pixels per meter. For *Marvel’s Spider-Man* impostor zones were a combination of up to 9 tile regions and the atlases for each individual impostor were collected into a single 8k atlas where the spec, gloss and reflective properties were scaled down 50%. By comparison, the impostor zones for *Marvel’s Spider-Man: Miles Morales* on the PlayStation 5 represented a single tile region without down sampling or resizing any of the components. An additional atlas type encoded the geometric normals used to add additional detail with bump functions [Blinn, 1978]. To support all the atlas types without resizing, the

atlases were combined into dynamic arrays of up to eight 4k textures using 2d rectangular bin packing [Jylänki, 2010] to optimize the placement of the individual building textures into the texture atlas array for a tile region.

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